Team and Setup

Software Engineering, Fall 2026

Project Theme: Environment and Sustainability

1. Team Information

Field	Details
Team Name	FareShare
Section	102112
Member 1	Uroos Fatima - 31094
Member 2	Aleena Aqeel - 31635
Member 3	Muhammad Shayan Hussain - 30077
Member 4	Waniya Jabeen - 31624

2. Project Type and Theme

- Project Type: Mobile Application
- Theme: Environment and Sustainability

3. Product Description

FareShare is a carpooling coordination app built for IBA students, designed to cut down on single-occupancy commuting and the resulting fuel consumption and emissions. Today, students coordinate carpools through scattered WhatsApp groups (e.g. “carpooling - gulshan”, “carpooling - dha”), where requests get buried and there is no reliable way to confirm a match. FareShare replaces this with structured trip posting, search, and confirmation, restricted to verified IBA students, so that more commutes are shared rather than driven solo, directly supporting the environment and sustainability goals of the project.

4. Target User and Problem

- Target user: verified IBA students (signup restricted to @khi.iba.edu.pk email addresses) who commute regularly to IBA's Main Campus or City Campus from the same residential areas (e.g. Gulshan, DHA) and currently rely on informal, area-based WhatsApp carpooling groups.
- Problem: WhatsApp groups do not scale, trip announcements get buried in chat history, there is no way to filter by route, timing, or vehicle type, and students often send a request and simply wait with no certainty of a response. There is also no way to know who you would be traveling with beyond a text message. This uncertainty, combined with the lack of a structured way to confirm a match, discourages carpooling and keeps more students defaulting to solo trips, working against the environment and sustainability goal of reducing unnecessary individual vehicle use.

5. Structural Minimums

At least 3 CRUD entities

- Trips: a trip manager creates, edits, cancels, and views trip postings (route/area, date & time, trip type: own car or shared InDrive, seats needed, cost-split method).
- Trip Requests: a companion creates a request to join a trip; the trip manager and companion can view, update (accept/decline), or cancel the request as it moves toward confirmation.
- Vehicles: a trip manager adds, edits, removes, and views their registered vehicle(s) (make, model, plate, seat capacity), relevant when a trip uses an own car.

4 core features

- Account and Profile: Login, Logout, Signup/Register (restricted to @khi.iba.edu.pk emails), Forgot Password, Change Password, and Profile, including gender and a completed-trips count, bundled as 1 feature.
- Trip Posting and Management: a trip manager posts, edits, or cancels a trip, specifying route/area, timing, trip type (own car vs. needs shared InDrive), seat availability, and cost-split method (fare or petrol split).
- Search and Request Trips: companions filter nearby trips by area, timing, trip type, and gender preference, then send a request to join.
- Trip Confirmation Coordination: the multi-step transactional flow (see below).

Multi-step transactional flow

- Step 1: Companion selects a trip and sends a join request.
- Step 2: Trip manager reviews the request and accepts or declines it.
- Step 3: If the trip uses an own car, the trip manager confirms the exact pickup point and time with accepted companions. If the trip needs a shared InDrive, the trip manager arranges the booking once enough companions are confirmed and shares the pickup point and time with the group.
- Step 4: The trip manager marks the trip complete, recording the fare/petrol split and incrementing each participant's completed-trips count.

6. Tool Setup Confirmation

- Trello board created and shared: https://trello.com/invite/b/6aad7fd0d610eccc91520f1e/ATTI2b19e6e15a411a3b471ad0a4d8b7bacf124DA912/fareshare
- Repository created and shared (GitHub/GitLab/Bitbucket): FareShare GitHub